

Unit 4

Instrumentation techniques in chemical analysis

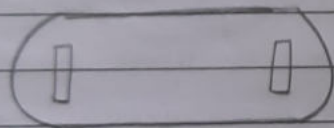
Spectroscopy

study of spectrum → Representation of radiation
→ Study of interaction b/w energy & matter

Visible Hydrogen

possible ↓ because of
Quantisation of energy

Hydrogen spectrum

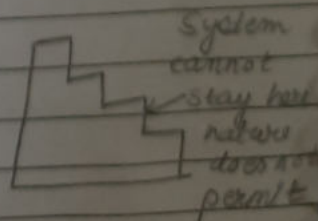


Electrodes are used to maintaining potential difference (above 10K)

The e^- first gets excited and then realised that it is now unstable. It comes back down giving out radiation. This radiation forms emission spectrum

Ques Why is CO_2 a greenhouse gas?
 CO_2 absorbs radiation when the e^- comes back down, heat is absorbed & released.

★ Energy is always quantised $E = h\nu$



Limitations of microwave oven

can only heat things with moisture.

Type of energy level $\rightarrow$ characteristic feature of that metal

Water molecule
 $\downarrow$
 Radiation lega
 $\downarrow$
 excite hoga
 $\downarrow$
 Heat emit karke
 $\downarrow$
 Relax hoga

Electronic

Vibrational

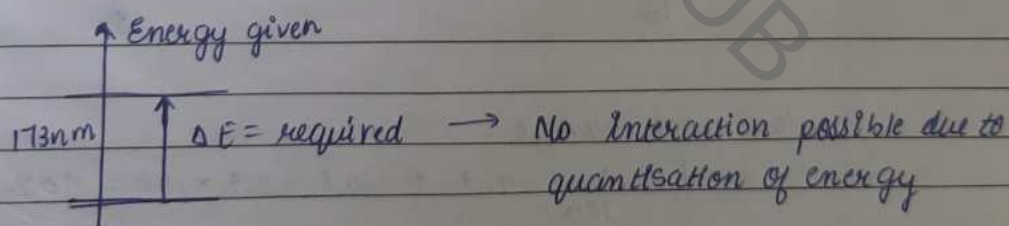
Rotational

Nuclear magnetic

Why can't we cook potatoes using IR.



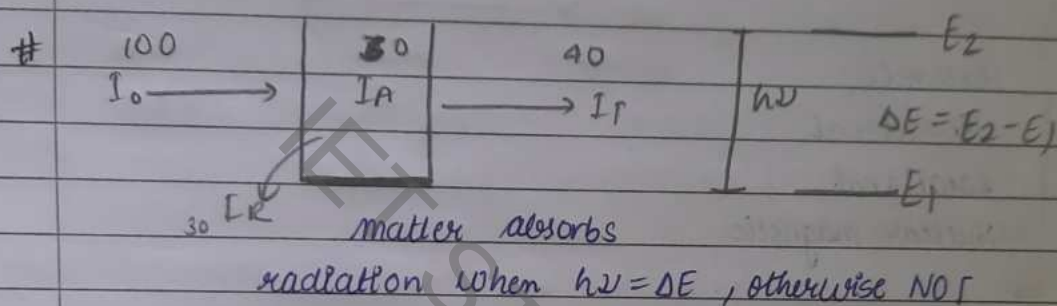
(Phone emitting IR)



★ Instrumentation techniques

- $\rightarrow$ No requirement of large amounts of sample
- $\rightarrow$ Sample is reusable
- $\rightarrow$ Time taken is less (1-2 min)
- $\rightarrow$ Higher accuracy, once calibrated — give uniform results not dependant on environmental conditions

- Reproducible
- Instruments are always more sensitive, hence give more reliable results.
- Less chances of side reactions / Does not show results based on chemical purity since no chemical is used.
- The entire process can be automated



$$I = I_{\text{lost}(R)} + I_{\text{Absorbed}(A)} + I_{\text{transmit}(T)}$$

Transmittance (T) → Dimensionless

$$T = \frac{I_T}{I_0} \quad \text{and} \quad \%T = T \times 100 = \frac{I_T}{I_0} \times 100$$

for above example:

$$T = \frac{40}{100} = 0.4 \quad \& \quad \%T = 0.4 \times 100 = 40\%$$

Absorbance

$$A = -\log T = \log\left(\frac{1}{T}\right)$$

$$= -\log \frac{\%T}{100} = \log \frac{100}{\%T}$$

$$= \log 100 - \log \%T$$

$$A = 2 - \log \%T$$

This is also known as optical density.

Ques Calculate absorbance value if sample absorbs 90% light

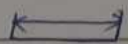
$$\%T = 10\% = 100 - \%A$$

$$A = 2 - \log 10$$
$$= 2 - 1 = 1$$

If radiation characteristic is constant. Interactions are increased by:

Increasing concentration

Increase path length



To find the ratio in which interactions increase with path length, experiments were conducted by Lambert

Lambert's law

When monochromatic light passes through transparent medium, the change in intensity of incident light w.r.t the thickness of the medium is proportional to intensity of incident light.

Date

$$-\frac{dI}{dx} \propto I$$

$$\frac{dI}{dx} = -KI$$

$$dI = -Kdx$$

$$\int_{I_0}^{I_1} \frac{dI}{I} = -K \int_0^x dx$$

$$[\ln I]_{I_0}^{I_1} = -K[x]_0^x$$

$$\ln I_1 - \ln I_0 = -K(x-0)$$

$$\ln \frac{I_1}{I_0} = -Kx$$

$$\frac{I_1}{I_0} = e^{-Kx}$$

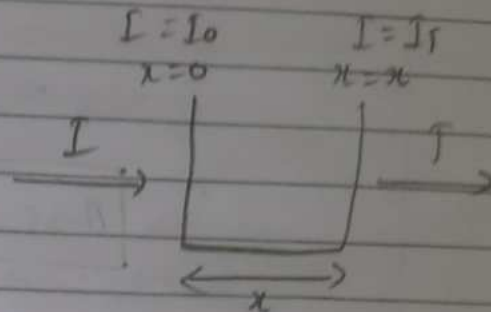
$$I_1 = I_0 e^{-Kx}$$

$$2.303 \log \frac{I_1}{I_0} = -Kx$$

$$\log \frac{I_1}{I_0} = \left(\frac{-K}{2.303} \right) \cdot x$$

$$-\log \left(\frac{I_1}{I_0} \right) = K'x$$

$$A \propto x$$



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Beer's law

When monochromatic light passes through transparent medium, the intensity of incident light decreases exponentially as the concentration of the medium increases arithmetically.

$A \propto c \rightarrow$ From Lambert's law

$A \propto \lambda$

$A \propto \lambda \cdot c$

$A = \epsilon \times c \rightarrow \text{mol/L}$

ϵ = molar absorption coefficient

λ = path length (conventionally cm)

c = concentration (mol/L)

[In ex cm, it will be in m]

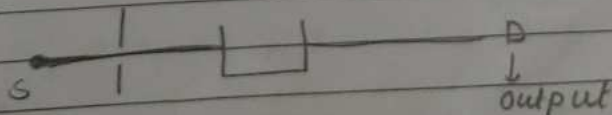
A = unitless

Colorimetry

Objective $\rightarrow$ To determine concentration of unknown coloured sample.

Limitations $\rightarrow$

- * Not applicable for colourless sample.
- * It is used for dilute solutions.



400-800 nm

Date

$$A = \epsilon \times C$$

$$A_1 = \epsilon \times C_1$$

$$A_2 = \epsilon \times C_2$$

$$\epsilon = \frac{A_1}{C_1}, \quad \epsilon = \frac{A_2}{C_2}$$

$$\boxed{\frac{A_1}{C_1} = \frac{A_2}{C_2}} \Leftrightarrow \boxed{\frac{A_1}{A_2} = \frac{C_1}{C_2}}$$

$$A = \epsilon \times C + 0$$

$$y = mx + c$$

$$C_1 \rightarrow \text{---} A_1$$

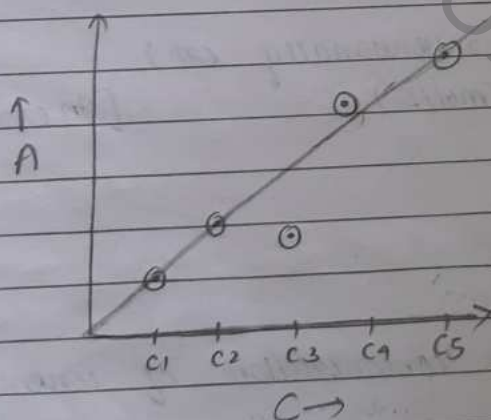
$$C_2 \rightarrow \text{---} A_2$$

$$C_3 \rightarrow \text{---} A_3$$

$$C_4 \rightarrow \text{---} A_4$$

$$C_5 \rightarrow \text{---} A_5$$

Sample $\rightarrow$ $\text{KMnO}_4 + \text{H}_2\text{O}$



(calibration curve method)

Prepare a stock soln. $1 \times 10^{-4} \text{ M}$

5ml	10ml	15ml	20ml	25ml
95ml	90ml	85ml	80ml	75ml
distilled				
water				

$$M_1 V_1 = M_2 V_2$$

$$1 \times 10^{-4} \times 5 = M_2 \times 100$$

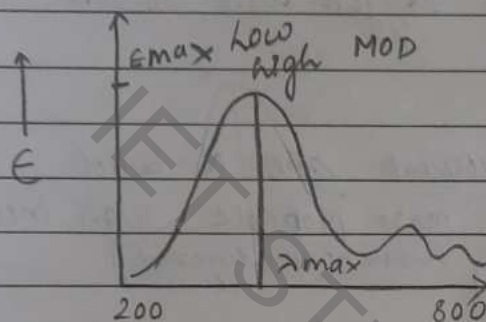
$$M_2 = \frac{1 \times 10^{-4} \times 5}{100}$$

Similarly for all #

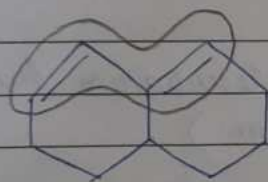
To find $\epsilon \rightarrow$ First prepare a known sol.

★ UV - vis spectroscopy, (200-800nm) (Electronic spectroscopy)

This technique is used for the detection of double bonds unsaturation.

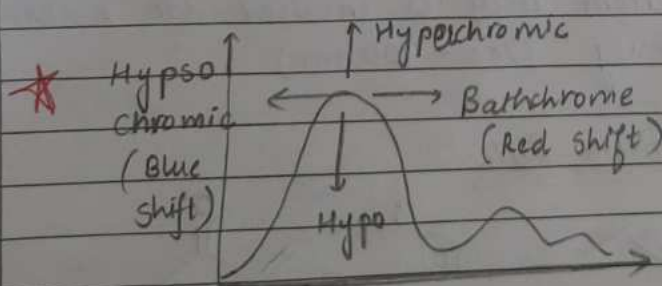


Auxochrome
(which can change absorption properties / intensity & position of chromophore)



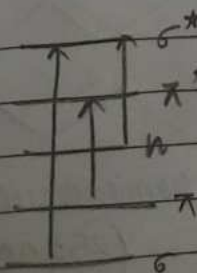
Chromophore

$C=C$ $N=N$
 $C=N$
 $C=O$
 $C=S$



Condition $\rightarrow$

There must be atleast 1 π bond



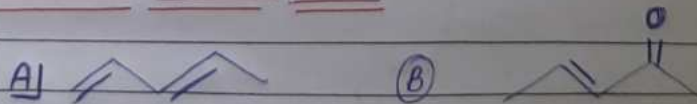
1. $\sigma \rightarrow \sigma^*$
 - Very low Intensity
 - For saturated H.C. transition
 - High amount of energy is req. as band gap is high
 - transition is seen below 100 nm (vacuum UV region) because otherwise Oxygen will absorb.

2. $\pi \rightarrow \pi^*$
 - There must be atleast one π -bond
 - Most favourable, most probable, high intensity transition (as less energy is req) (weak bonds, delocalised π)
200 nm - 700 nm

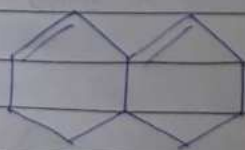
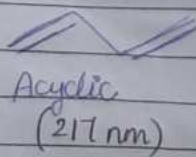
3. $n \rightarrow \sigma^*$
 - There must be atleast one heteroatom having lone pair
 - low intensity (150 nm - 250 nm)

4. $n \rightarrow \pi^*$
 - when hetero atom itself is involved in π -bond
 - Moderate intensity (200 - 700 nm)

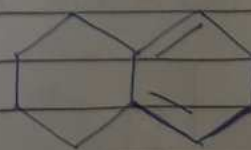
* Woodward Fieser Rule



A)  Conjugated diene



Heteroannular
(214 nm)



Homocannular
(253 nm)

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→ For each extra double bond = +30

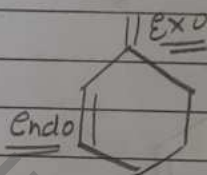


$BV = 217$
 $EDB = +60$

→ Exocyclic DB = +5

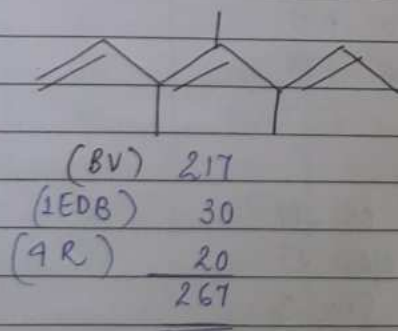
→ R = +5

→ Ring Residual = +5



(one C is connected to other C which is not part of ring)

Ans



→ OH = +6

→ OR = +6

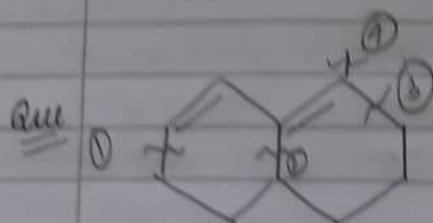
→ Cl = +5

→ -SH = +30

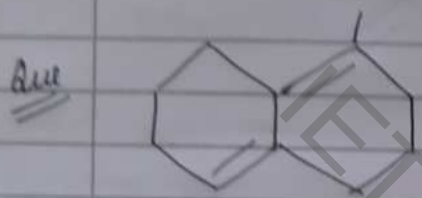
→ -NH₂ = +60

→ -OCOR = +0

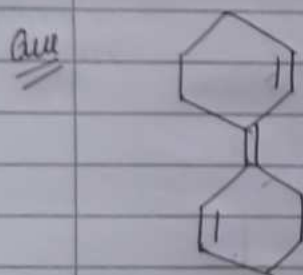
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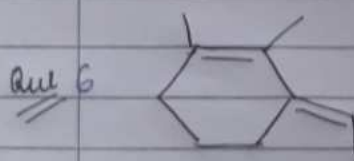
$$\begin{aligned} \text{BV} & 214 \\ \text{R/Rs} & + 20 \\ \text{Exo} & + 5 \\ \hline & 239 \text{ nm} \end{aligned}$$



$$\begin{aligned} \text{BV} & 214 \\ \text{R/Rs} & + 25 \\ \text{Exo} & + 10 \\ \hline & 249 \end{aligned}$$



$$\begin{aligned} \text{BV} & 217 \\ \text{R/Rs} & + 20 \\ \text{Exo} & + 10 \\ \text{ExDB} & + 30 \\ \hline & 277 \text{ nm} \end{aligned}$$

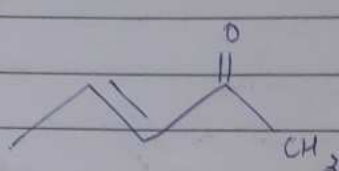


$$\begin{aligned} \text{BV} & 217 \\ \text{R/Rs} & 25 \\ \text{Exo} & 5 \\ \hline & 247 \text{ nm} \end{aligned}$$

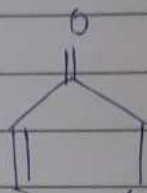
8

$$\begin{aligned} & 217 \\ & 10 \\ & 30 \\ \hline & 257 \text{ nm} \end{aligned}$$

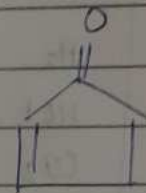
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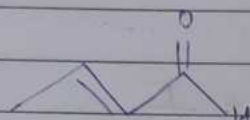
215



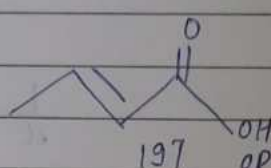
215



202



207



197

OH
OR

Extra DB = +30

Exocyclic = +5

Homodiene component = +39

	α	β	γ	δ
R	10	12	18	18
OH	35	30	30	50
-OR	35	30	17	31
-OCOR	6	6	6	
Cl	15	12		
Br	25	30		

IR Spectroscopy

Range $\rightarrow (1200 - 1000) \text{ cm}^{-1}$

Usable range $\rightarrow (4000 - 400) \text{ cm}^{-1}$

CO_2 (3N-5) (Trans-3, Rot-2)

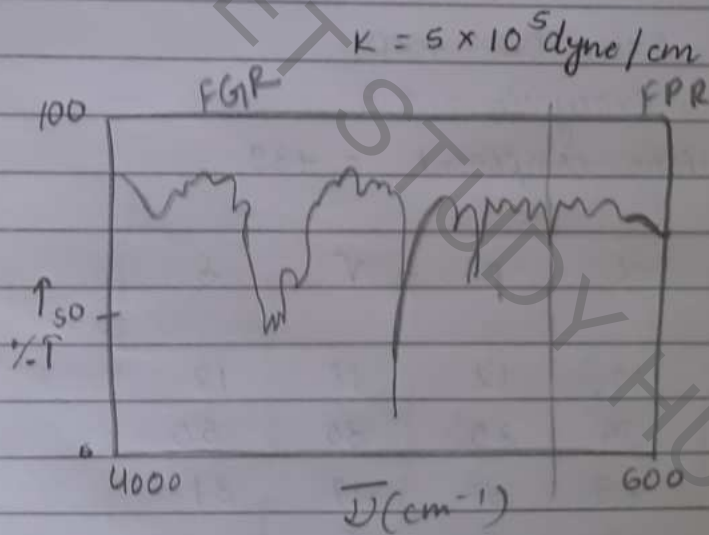
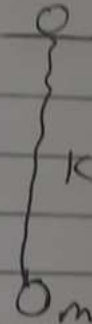
H_2O (3N-6) (Trans-3, Rot-3)

Date

H₂
HCl
CO₂
H₂O
NH₃
CH₄
C₆H₆

$$\bar{\nu} = \frac{\nu}{c} = \frac{1}{2\pi c} \sqrt{\frac{k}{m}}$$

$$\frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2}$$



OH (3800-3500)